

Nepal PreTST 2025 Solutions

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§1 Problems

Problem 1.1. Shining tells Prajit a positive integer $n \geq 2025$. Prajit then tries to place n points such that no four points are concyclic and no 3 points are collinear in Euclidean plane, such that Shining cannot find a group of three points such that their circumcircle contains none of the other remaining points. Is he always able to do so?

Proof. □

Problem 1.2. (a) Positive rational numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

(b) Positive real numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

Proof. □

Problem 1.3. Let the incircle of $\triangle ABC$ touch sides BC , CA , and AB at points D , E , and F , respectively. Let D' be the diametrically opposite point of D with respect to the incircle. Let lines AD' and AD intersect the incircle again at X and Y , respectively. Prove that the lines DX , $D'Y$, and EF are concurrent, i.e., the lines intersect at the same point.

Proof. □

Problem 1.4. Find all pairs of positive integers n and x such that

$$1^n + 2^n + 3^n + \cdots + n^n = x!$$

Proof. □